

Multiple colouring of cone graphs

Zhishi Pan and Xuding Zhu*

Department of Applied Mathematics

National Sun Yat-sen University

Kaohsiung, Taiwan 80424

and

National Center of Theoretical Sciences

Email: zhu@math.nsysu.edu.tw

Abstract

The k th chromatic number $\chi_k(G)$ of a graph G is the minimum number of colours needed so that each vertex can be assigned a set of k colours in such a way that colour sets assigned to adjacent vertices are disjoint. Given a graph $G = (V, E)$ and an integer $m \geq 0$, the m -cone of G , denoted by $\mu_m(G)$, has vertex set $(V \times \{0, 1, \dots, m\}) \cup \{u\}$ in which u is adjacent to every vertex of $V \times \{m\}$, and $(x, i)(y, j)$ is an edge if $xy \in E$ and $i = j = 0$ or $xy \in E$ and $|i - j| = 1$. This paper studies the k th chromatic number of the cone graphs. An upper bound for $\chi_k(\mu_m(G))$ in terms of $\chi_k(G)$, k and m is given. In particular, it is proved that for any graph G , if $m \geq 2k$, then $\chi_k(\mu_m(G)) \leq \chi_k(G) + 1$. We also find a surprising connection between the k th chromatic number of the cone graph of G and the circular chromatic number of G . It is proved that if $\chi_k(G)/k > \chi_c(G)$ and $\chi_k(G)$ is even, then for sufficiently large m , $\chi_k(\mu_m(G)) = \chi_k(G)$.

1 Introduction

Multiple colouring of graphs was introduced by Stahl in [13], and has been studied extensively in the literature. Suppose $n \geq k$ are positive integers. We denote by $[n]$ the set $\{0, 1, \dots, n-1\}$ and denote by $\binom{[n]}{k}$ the family of all k -subsets of $[n]$. Given a graph G and positive integers k, n , a k -fold n -colouring of G is a mapping $f : V \rightarrow \binom{[n]}{k}$ such that for any edge xy of G , $f(x) \cap f(y) = \emptyset$. In other words, a k -fold colouring assigns to each vertex a set of k colours, and no colour is assigned to two adjacent vertices. If all the colours are taken from

a set of n colours, then it is a k -fold n -colouring. If $k = 1$, then each vertex is assigned one colour, and adjacent vertices are assigned distinct colours. So a 1-fold n -colouring of G is simply an n -colouring of G . The k th chromatic number of G is defined as $\chi_k(G) = \min\{n : G \text{ has a } k\text{-fold } n\text{-colouring}\}$. As a 1-fold n -colouring is just an n -colouring, we have $\chi_1(G) = \chi(G)$. So $\chi_k(G)$ is a generalization of $\chi(G)$. It is well-known [11] and easy to see that for any $k, k' \geq 1$, $\chi_{k+k'}(G) \leq \chi_k(G) + \chi_{k'}(G)$. This implies that $\chi_k(G) \leq k\chi(G)$. The fractional chromatic number of G is defined as

$$\chi_f(G) = \inf\left\{\frac{\chi_k(G)}{k} : k = 1, 2, \dots\right\}.$$

Thus $\chi_f(G) \leq \chi(G)$.

It is known that the gap $\chi(G) - \chi_f(G)$ can be arbitrarily large. However, as observed in [10], only a few types of graphs G are known to have arbitrarily large gaps between $\chi(G)$ and $\chi_f(G)$. One class of such graphs are the Kneser graphs. Given positive integers $n \geq k$, the Kneser graph $K(n, k)$ has vertex set $\binom{[n]}{k}$ in which $A \sim B$ if and only if $A \cap B = \emptyset$. It is known [11] that for $n \geq 2k$, $\chi_f(K(n, k)) = n/k$. On the other hand, by Lovász Theorem [8], $\chi(K(n, k)) = n - 2k + 2$.

Another class of graphs G for which the gap $\chi(G) - \chi_f(G)$ are known to be arbitrarily large are obtained by cone constructions. Suppose $G = (V, E)$ is a graph and $m \geq 0$ is an integer. The m -cone $\mu_m(G)$ of G is the graph with vertex set $(V \times [m+1]) \cup \{u\}$ in which $(x, i)(y, j)$ is an edge if $xy \in E$ and $|i - j| = 1$ or $xy \in E$ and $i = j = 0$. Moreover, the vertex u is adjacent to (x, m) for all $x \in V$. If $m = 0$, then $\mu_m(G)$ is obtained from G by adding a universal vertex u . If $m = 1$, then the graph $\mu_m(G)$ is also called the Mycielskian of G and is denoted by $\mu(G)$. In the literature, the cone graph $\mu_m(G)$ for arbitrary integer $m \geq 0$ is also

*Supported in part by the National Science Council under grant NSC94-2115-M-110-001

called the generalized Mycielskian of G . Figure 1 is the 3-cone of C_5 .

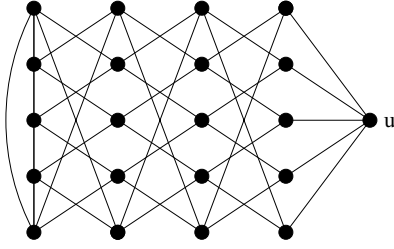


Figure 1: The graph $\mu_3(C_5)$.

The fractional chromatic number of $\mu_m(G)$ has been studied by Larsen, Propp and Ullman [9] and Tardif [14]. It is proved in [9] that for any graph G ,

$$\chi_f(\mu(G)) = \chi_f(G) + \frac{1}{\chi_f(G)}.$$

This result is generalized in [14], where it is proved that for any graph G ,

$$\chi_f(\mu_m(G)) = \chi_f(G) + \frac{1}{\sum_{i=0}^m (\chi_f(G) - 1)^i}.$$

We define $\mu_{m_1, m_2, \dots, m_t}(G)$ recursively as $\mu_{m_1, m_2, \dots, m_t}(G) = \mu_{m_1}(\mu_{m_2, m_3, \dots, m_t}(G))$. Then by choosing the integers m_i to be sufficiently large, the formula above shows that $\chi_f(\mu_{m_1, m_2, \dots, m_t}(K_2))$ is “close” to 2. On the other hand, it is known [3] that $\chi(\mu_{m_1, m_2, \dots, m_t}(K_2)) = 2 + t$. So for $G = \mu_{m_1, m_2, \dots, m_t}(K_2)$, the gap $\chi(G) - \chi_f(G)$ is close to t . Indeed, for any given graph H , it is known [12] that for $m_i \geq 1$ for $i \geq 1$, for $G_0 = H$ and $G_t = \mu_{m_t}(G_{t-1})$, $\lim_{t \rightarrow \infty} (\chi(G_t) - \chi_f(G_t)) = \infty$, although the exact value of the gap $\chi(G_t) - \chi_f(G_t)$ might be difficult to determine.

For both the Kneser graphs and the cone graphs, the determination of their chromatic numbers is non-trivial. The problem of determining the chromatic number of Kneser graphs remained an open problem for more than 20 years before it was settled by Lovász by an ingenious application of algebraic topology. The proof in [3] of the fact that $\chi(\mu_{m_1, m_2, \dots, m_t}(K_2)) = 2 + t$ also uses topological method.

As the k th chromatic number of graphs is a generalization of the chromatic number, it is natural to investigate the k th chromatic numbers of these graphs. The problem of determining the k th chromatic number of Kneser graphs, which is equivalent to the problem of determining existence of

homomorphisms among Kneser graphs, has been studied in the literature [2, 4, 13]. It is known that if $k = qk' - r$, where $q \geq 1$ and $0 \leq r < k'$ are integers, then $\chi_k(K(n', k')) \leq qn' - 2r$. It was conjectured in [13] that equality holds. The conjecture is verified for $r = 0$ or $q = 1$, but open for other cases.

In this paper, we are interested in the k th chromatic number of cone graphs. As $\mu_m(G)$ contains G as a subgraph, we have $\chi_k(\mu_m(G)) \geq \chi_k(G)$. On the other hand, given a k -fold colouring f of G , we obtain a k -fold colouring of $\mu_m(G)$ by letting $g(x, i) = f(x)$ for $x \in V(G)$ and for $i = 0, 1, \dots, m$, and letting $g(u)$ be a set of k new colours. So for any graph G , for any $m \geq 0$, for any $k \geq 1$,

$$\chi_k(G) \leq \chi_k(\mu_m(G)) \leq \chi_k(G) + k.$$

Both the upper and lower bounds can be attained. Indeed, for any graph G , for any integer $k \geq 1$, $\chi_k(\mu_0(G)) = \chi_k(G) + k$, as $\mu_0(G) - u$ is a copy of G , and the k colours assigned to the vertex u cannot be assigned to any other vertex of $\mu_0(G)$. However, if m is large then the upper bound can be improved. We shall prove an upper bound for $\chi_k(\mu_m(G))$ in terms of $\chi_k(G)$, k and m . In particular, the upper bound implies that if $m \geq 2k$, then $\chi_k(\mu_m(G)) \leq \chi_k(G) + 1$. This upper bound is the best in general, because if $\chi_k(G) = k\chi_f(G)$, then it follows from the above mentioned result of Tardif that for any m , $\chi_k(\mu_m(G)) \geq k\chi_f(\mu_m(G)) > k\chi_f(G) = \chi_k(G)$. However, for some graphs G and integers k , the upper bound can be further improved. Surprisingly, the k th chromatic number of $\mu_m(G)$ is related to the circular chromatic number of G . We prove that if $\chi_k(G)/k > \chi_c(G)$ and $\chi_k(G)$ is even, then for sufficiently large m , we have $\chi_k(\mu_m(G)) = \chi_k(G)$. If $\chi_c(G) = p/q$ and p is odd, then for any $k \geq q$, if $\chi_k(G)/k > \chi_c(G)$, then for sufficiently large m , we have $\chi_k(\mu_m(G)) = \chi_k(G)$.

2 A general upper bound for $\chi_k(\mu_m(G))$

The m -cone of a graph G can be equivalently defined as follows: Given two graphs $G = (V, E)$ and $G' = (V', E')$. The *categorical product* $G \times G'$ of G and G' has vertex set $V \times V'$ in which $(x, x')(y, y')$ is an edge if $xy \in E$ and $x'y' \in E'$. We denote by P_m the path of length m with vertex set $\{0, 1, \dots, m\}$, and denote by P_m^* the graph

obtained from P_m by adding a loop on vertex 0. Then $\mu_m(G)$ is the graph obtained from $G \times P_m^*$ by adding a vertex u which is adjacent to every vertex of $V \times \{m\}$. We call a k -fold n -colouring of $G \times P_m^*$ an *extendable k -fold n -colouring* if the restriction of f to $V \times \{m\}$ uses at most $n - k$ colours. The following lemma follows easily from the the definitions.

Lemma 1 *For any graph $G = (V, E)$ and integers k, m , $\chi_k(\mu_m(G)) \leq n$ if and only if there is an extendable k -fold n -colouring of $G \times P_m^*$.*

Lemma 2 *For any graph G , for any integers $m \geq 0, k \geq 1$,*

$$\chi_k(\mu_{m+1}(G)) \leq \chi_k(\mu_m(G)).$$

Proof. Assume $\chi_k(\mu_m(G)) = n$. By Lemma 1, there is an extendable k -fold n -colouring f of $G \times P_m^*$. Let B be a k -subset of $\{0, 1, \dots, n - 1\}$ that are colours not used by $V(G) \times \{m\}$. Then the mapping ϕ defined as $\phi(x, i) = f(x, i)$ for $i = 0, 1, \dots, m$ and $\phi(x, m + 1) = B$ is an extendable k -fold n -colouring of $G \times P_{m+1}^*$. ■

The following theorem gives an upper bound for $\chi_k(\mu_m(G))$ in terms of $\chi_k(G), k$ and m .

Theorem 3 *For any finite graph G , for any positive integers m, k, s , if $m \geq 2\lceil k/s \rceil$, then $\chi_k(\mu_m(G)) \leq \chi_k(G) + s$*

Proof. Assume $\chi_k(G) = n$. Let ϕ be a k -fold n -colouring of G , using colour set $[n] = \{0, 1, \dots, n - 1\}$. We shall construct an extendable k -fold $(n + s)$ -colouring ψ of $G \times P_m^*$ using colour set $\{c_1, c_2, \dots, c_s, 0, 1, \dots, n - 1\}$.

Subsets of $\{0, 1, \dots, n - 1\}$ will be denoted by capital letters $A, B, \dots$. The elements of a k -element set A will be denoted as $A = \{a_1, a_2, \dots, a_k\}$ in such a way that $a_1 < a_2 < \dots < a_k$, and elements of a k -element set B will be denoted as $B = \{b_1, b_2, \dots, b_k\}$ with $b_1 < b_2 < \dots < b_k$, etc. The complement $[n] \setminus A$ of A is denoted by $\bar{A} = \{\bar{a}_1, \bar{a}_2, \dots, \bar{a}_{n-k}\}$ where $\bar{a}_1 < \bar{a}_2 < \dots < \bar{a}_{n-k}$.

Assume $k = qs + r$, where $0 \leq r \leq s - 1$. By Lemma 2, it suffices to consider the case that $m = 2\lceil k/s \rceil = 2q$. We construct an extendable k -fold colouring of $G \times P_{2q}^*$ as follows: Suppose $x \in V$ and $\phi(x) = A = \{a_1, a_2, \dots, a_k\}$. Then $\psi(x, 0) = \phi(x)$. For $1 \leq i \leq q$, let

$$\psi(x, 2i - 1) = \{c_1, c_2, \dots, c_s, 0, 1, \dots, (i - 1)s - 1\} \cup \{a_{is+1}, a_{is+2}, \dots, a_k\}$$

and

$$\psi(x, 2i) = \{a_{t+1}, a_{t+2}, \dots, a_k\} \cup \{\bar{a}_{is-t+1}, \bar{a}_{is-t+2}, \dots, \bar{a}_{is}\},$$

where $t = |A \cap \{0, 1, \dots, is - 1\}|$.

Now we shall prove that ψ is indeed an extendable k -fold $(n + s)$ -colouring of $G \times P_m^*$. It follows easily from the definition that for any $x \in V$ and for any $0 \leq i \leq m$, $\psi(x, i)$ is a k -subset of $[n] \cup \{c_1, c_2, \dots, c_s\}$. Assume $(x, j) \sim_{G \times P_m^*} (y, j')$. Then either $j = j' = 0$ and $x \sim_G y$, or $j' = j + 1$ and $x \sim_G y$. We need to show that $\psi(x, j) \cap \psi(y, j') = \emptyset$. Assume $\phi(x) = A$ and $\phi(y) = B$. As ϕ is a k -fold colouring of G , $A \cap B = \emptyset$. If $j = j' = 0$, then $\psi(x, j) = A$ and $\psi(y, j') = B$ and we are done. Assume $j' = j + 1$. If $j = 2i$ is even, then

$$\psi(x, 2i) = \{a_{t+1}, a_{t+2}, \dots, a_k\} \cup \{\bar{a}_{is-t+1}, \bar{a}_{is-t+2}, \dots, \bar{a}_{is}\}$$

where $|A \cap \{0, 1, \dots, is - 1\}| = t$ and

$$\psi(y, 2i + 1) = \{c_1, c_2, \dots, c_s, 0, 1, \dots, is - 1\} \cup \{b_{(i+1)s+1}, b_{(i+1)s+2}, \dots, b_k\}.$$

To prove that $\psi(x, 2i) \cap \psi(y, 2i + 1) = \emptyset$, it suffices to show that

$$\begin{aligned} \{a_{t+1}, a_{t+2}, \dots, a_k\} \cap \{b_{(i+1)s+1}, b_{(i+1)s+2}, \dots, b_k\} &= \emptyset. \\ \{a_{t+1}, a_{t+2}, \dots, a_k\} \cap \{c_1, c_2, \dots, c_s, 0, 1, \dots, is - 1\} &= \emptyset. \\ \{\bar{a}_{is-t+1}, \bar{a}_{is-t+2}, \dots, \bar{a}_{is}\} \cap \{b_{(i+1)s+1}, b_{(i+1)s+2}, \dots, b_k\} &= \emptyset. \\ \{\bar{a}_{is-t+1}, \bar{a}_{is-t+2}, \dots, \bar{a}_{is}\} \cap \{c_1, c_2, \dots, c_s, 0, 1, \dots, is - 1\} &= \emptyset. \end{aligned}$$

The first equality holds because $A \cap B = \emptyset$. The second equality holds because $|A \cap \{0, 1, \dots, is - 1\}| = t$, which implies that $a_{t+1} \geq is$. The third equality holds because $B \subseteq \bar{A}$, which implies that $\bar{a}_{is} \leq b_{is} < b_{(i+1)s+1}$. The fourth equality holds because $\bar{A} \cap \{0, 1, \dots, is - 1\} = is - t$, which implies that $\bar{a}_{is-t+1} \geq is$.

Assume $j = 2i - 1$ is odd. Then

$$\psi(x, 2i - 1) = \{c_1, c_2, \dots, c_s, 0, 1, \dots, (i - 1)s - 1\} \cup \{a_{is+1}, a_{is+2}, \dots, a_k\}$$

and

$$\psi(y, 2i) = \{b_{t+1}, b_{t+2}, \dots, b_k\} \cup \{\bar{b}_{is-t+1}, \bar{b}_{is-t+2}, \dots, \bar{b}_{is}\},$$

where $t = |A \cap \{0, 1, \dots, is - 1\}|$.

Again, since $|B \cap \{0, 1, \dots, is - 1\}| = t$ we have $\bar{B} \cap \{0, 1, \dots, is - 1\} = is - t$. Therefore $b_{t+1} \geq is$

and $\bar{b}_{is-t+1} \geq is$. Since $A \subseteq \bar{B}$, we have $\bar{b}_{is} \leq a_{is} < a_{is+1}$. Similarly as above, this implies that $\psi(x, 2i-1) \cap \psi(y, 2i) = \emptyset$.

Moreover, the colours in the set $\{c_1, c_2, \dots, c_s\} \cup \{0, 1, \dots, qs-1\}$ are not used in $V \times \{2q\}$. So ψ is indeed an extendable k -fold $(n+s)$ -colouring of $G \times P_m^*$, and hence $\chi_k(\mu_m(G)) \leq n+s$. ■

Corollary 4 *For any integer $k \geq 1$ and for any graph $G = (V, E)$, if $m \geq 2k$, then $\chi_k(\mu_m(G)) \leq \chi_k(G) + 1$.*

For the Mycielskian $\mu(G)$ of a graph G , the upper bound given in Theorem 3 is not better than the trivial bound $\chi_k(\mu(G)) \leq \chi_k(G) + k$. The following theorem shows that if $\chi_k(G) \leq 3k-2$, then we do have a better bound.

Theorem 5 *Suppose G is a graph with $\chi_k(G) = n \leq 3k-2$. Then $\chi_k(\mu(G)) \leq n+k-1$.*

Proof. Let ϕ be a k -fold n -colouring of G , using colours $\{0, 1, \dots, n-1\}$. We shall construct a k -fold $(n+k-1)$ -colouring f of $\mu(G)$. Suppose $x \in V(G)$ and $\phi(x) = \{a_1, a_2, \dots, a_k\} \in \binom{[n]}{k}$, where $a_1 < a_2 < \dots < a_k$. Let

$$f(x, 0) = \begin{cases} \phi(x), & \text{if } a_1 = 0 \\ \{a_2, \dots, a_k, n+a_1-1\}, & \text{if } 1 \leq a_1 \leq n-2k \\ \{a_2, \dots, a_k, n+k-2\}, & \text{if } a_1 \geq n-2k+1. \end{cases}$$

$$f(x, 1) = \begin{cases} \phi(x), & \text{if } 0 \notin \phi(x) \\ (\phi(x) \setminus \{0\}) \cup \{\bar{a}_1\}, & \text{otherwise.} \end{cases}$$

$$f(u) = \{0, n, n+1, \dots, n+k-2\}.$$

The argument similar to the proof of Theorem 3 shows that f is a k -fold $(n+k-1)$ -colouring of $\mu(G)$. We omit the details. ■

3 Graphs with $\chi_f(G) = \chi_c(G)$

Theorem 3 provides an upper bound for $\chi_k(\mu_m(G))$ in terms of $\chi_k(G)$, k and m for general graphs G . In this section, we show that if G has a k -fold colouring with some special properties, then there is a better upper bound for $\chi_k(\mu_m(G))$. In particular, we shall obtain a better upper bound for graphs G for which $\chi_f(G) = \chi_c(G)$.

Lemma 6 *Suppose f is a k -fold n -colouring of $G \times P_m^*$, using colours $\{0, 1, \dots, n-1\}$. Suppose X, Y are two sets of colours such that*

- X is a set of colours not used by any vertex in $V(G) \times \{m\}$, and

- for any $x \in V(G)$, $|f(x, m-1) \cap Y| \leq |X \setminus f(x, m-1)|$.

Then f can be extended to a k -fold n -colouring of $G \times P_{m+1}^$ such that no colour in Y is used by any vertex in $V(G) \times \{m\}$.*

Proof. For any $x \in V(G)$, since $|f(x, m-1) \cap Y| \leq |X \setminus f(x, m-1)|$, we can choose a set B of $|f(x, m-1) \cap Y|$ colours from $X \setminus f(x, m-1)$. Let

$$f(x, m+1) = (f(x, m-1) \setminus Y) \cup B.$$

Then it is straightforward to verify that the resulting mapping is a required extension of f to $G \times P_{m+1}^*$. ■

By repeated applying Lemma 6, we have the following corollary.

Corollary 7 *Suppose f is a k -fold n -colouring of $G \times P_m^*$, using colours $\{0, 1, \dots, n-1\}$. Assume there exist sets $X_1, \dots, X_t$ of colours such that the following hold:*

- Colours in X_1 are not used by any vertex in $V(G) \times \{m\}$.
- For $i \geq 1$, for $x \in V(G)$, $|f(x, m-1) \cap X_{2i}| \leq |X_{2i-1} \setminus f(x, m-1)|$, and $|f(x, m) \cap X_{2i+1}| \leq |X_{2i} \setminus f(x, m)|$.
- For $i = 1, 2, \dots, t-1$, $X_i \cap X_{i+1} = \emptyset$.
- For $i \geq 1$, $X_{2i-1} \subseteq X_{2i+1}$ and $X_{2i} \subseteq X_{2i+2}$.

Then there is a k -fold n -colouring ϕ of $G \times P_{m+t-1}^$ such that the restriction of ϕ to $V \times \{0, 1, \dots, m\}$ is equal to f , and $\phi(V(G) \times \{m+i-1\}) \cap X_i = \emptyset$. In particular, the colours in X_t are not used by any vertex in $V(G) \times \{m+t-1\}$.*

Proof. For each $x \in V(G)$ for $i \geq 1$, choose a set B of $|f(x, m-1) \cap X_{2i}|$ colour from $X_{2i-1} \setminus f(x, m-1)$, and let $f(x, m+2i-1) = (f(x, m-1) \setminus X_{2i}) \cup B$; choose a set C of $|f(x, m) \cap X_{2i+1}|$ colours from $X_{2i} \setminus f(x, m)$ and let $f(x, m+2i) = (f(x, m) \setminus X_{2i+1}) \cup C$. It is easy to verify that f is a k -fold colouring of $G \times P_{m+t-1}^*$, and $\phi(V(G) \times \{m+i-1\}) \cap X_i = \emptyset$ for $i = 1, 2, \dots, t$. ■

Corollary 8 *Suppose f is a k -fold n -colouring of a graph G . Assume there exist sets $X_1, \dots, X_m$ such that the following hold:*

- Colours in X_1 are not used by any vertex in $V(G)$.
- For $i = 1, 2, \dots$, $|f(x) \cap X_{i+1}| \leq |X_i \setminus f(x)|$.

- For $i = 1, 2, \dots, m-1$, $X_i \cap X_{i+1} = \emptyset$.
- For $i \geq 1$, $X_{2i-1} \subseteq X_{2i+1}$ and $X_{2i} \subseteq X_{2i+2}$.

Then there is a k -fold n -colouring ϕ of $G \times P_m^*$ such that for any $x \in V(G)$, $\phi(x, 0) = f(x)$ and $\phi(V \times \{m\})$ uses at most $n - |X_m|$ colours.

Proof. Since f is a k -fold n -colouring of G , by letting $f'(x, 0) = f'(x, 1) = f(x)$, we obtain a k -fold n -colouring of $G \times P_1^*$. Then the conclusion follows from Corollary 7. ■

Given positive integers $p \geq 2q \geq 2$. The graph $K_{p/q}$ has vertex set $\{0, 1, \dots, p-1\}$ in which ij is an edge if $q \leq |i-j| \leq p-q$. When $q = 1$, $K_{p/1}$ is the complete graph K_p . In general, the graphs $K_{p/q}$ are called the *circular complete graphs*. It is known [16, 1, 17] that $K_{p/q} \rightarrow K_{p'/q'}$ if and only if $p/q \leq p'/q'$. So the set $\{K_{p/q} : p/q \geq 2\} \cup \{K_1\}$ form an increasing chain under the order of graph homomorphism. This chain is used to define the circular chromatic number of graphs. Given a graph G , the *circular chromatic number* $\chi_c(G)$ of G is defined as

$$\chi_c(G) = \inf\{p/q : G \rightarrow K_{p/q}\}.$$

It is known [17] that for any graph G , $\chi_f(G) \leq \chi_c(G) \leq \chi(G)$ and $\chi_c(G) > \chi(G) - 1$. For circular complete graphs $K_{p/q}$, we have $\chi_f(K_{p/q}) = \chi_c(K_{p/q}) = p/q$ [17].

We view the set $\{0, 1, \dots, p-1\}$ as a circle, and denote it by $S(p)$. For $a, b \in S(p)$, $[a, b]_p$ denotes the interval of the circle $S(p)$ from a to b along the increasing direction. For example, $[2, 5]_8 = \{2, 3, 4, 5\}$ and $[5, 2]_8 = \{5, 6, 7, 0, 1, 2\}$. The *length* of an interval A is denoted by $\ell(A)$ which is defined to be the number of elements in A . We use $[x]_p$ denotes the remainder of x upon division of p , and let $|x|_p = \min\{[x]_p, [-x]_p\}$. Then it is easy to verify that $[a, b]_p = \{[a+i]_p : 0 \leq i \leq [b-a]_p\}$, and that the interval $[a, b]_p$ has length $\ell([a, b]_p) = [b-a]_p + 1$. We say two intervals $[a, b]_p$ and $[c, d]_p$ are *consecutive* if $[b+1]_p = [c]_p$.

Lemma 9 For any $p/q \geq 2$ and for any integer $k \geq 1$,

$$\chi_k(K_{p/q}) = \min\{n : nq \geq pk\}.$$

Proof. Suppose $\chi_k(K_{p/q}) = n$. Since $n = \chi_k(K_{p/q})/k \geq \chi_f(K_{p/q}) = p/q$, we have $nq \geq pk$. It remains to prove that if $nq \geq pk$, then $\chi_k(K_{p/q}) \leq n$.

To prove that $\chi_k(G) \leq n$, it suffices to construct a family of independent sets $I_0, I_1, \dots, I_{n-1}$

of G such that each vertex of G is contained in k of these independent sets. Indeed, if such a family of independent sets are found, then we assign colour j to all the vertices in I_j . As each vertex is contained in k of the I_j 's, each vertex receives k colours. Hence we obtain a k -fold n -colouring of G . Let $t^* = nq - pk \geq 0$, and let

$$I_0 = [0, q - t^* - 1]_p,$$

$$I_j = [jq - t^*, (j+1)q - t^* - 1]_p, \text{ for } j = 1, 2, \dots, n-1.$$

Each I_j is an interval of length at most q . Hence each I_j is an independent set. Moreover, I_j and I_{j+1} are consecutive intervals for $j = 0, 1, \dots, n-1$ (where summation in the indices are modulo n). As

$$\sum_{j=0}^{n-1} \ell(I_j) = kp,$$

so the union of the intervals $I_0, I_1, \dots, I_{n-1}$ winds around the circle $S(p)$ exactly k times. Thus each vertex of $K_{p/q}$ is contained in exactly k of the independent sets I_j . So $\chi_k(K_{p/q}) \leq n$. ■

Corollary 10 If G is a graph with $\chi_c(G) = p/q$, then for any positive integer k , $\chi_k(G) \leq \lceil kp/q \rceil$.

Proof. If $\chi_c(G) = p/q$, then $G \rightarrow K_{p/q}$ and hence $\chi_k(G) \leq \chi_k(K_{p/q})$. ■

Theorem 11 Suppose G is a graph with $\chi_c(G) = p/q$. Let $z = \lfloor p/q \rfloor$. Then for any positive integer s , if $s(z-1)^m \geq k$, then $\chi_k(\mu_m(G)) \leq \lceil pk/q \rceil + s$.

Proof. Since $\chi_c(G) = p/q$, we have $G \rightarrow K_{p/q}$. Thus it suffices to show that $\chi_k(\mu_m(K_{p/q})) \leq \lceil pk/q \rceil + s$.

Let $n = \lceil pk/q \rceil$. By Lemma 9, $\chi_k(K_{p/q}) = n$. Let f be the k -fold n -colouring of $K_{p/q}$ as defined in the proof of Lemma 9. It is easy to verify that if $X = [a, b]_n$ is a set of consecutive integers, then for any vertex $x \in V(K_{p/q})$, $|f(x) \cap X| \leq \lceil q|X|/p \rceil$. Hence $|X \setminus f(x)| \geq |X| - \lceil q|X|/p \rceil = \lfloor |X|(p-q)/p \rfloor \geq |X|(z-1)$. Assume m is an integer with $s(z-1)^m \geq k$. For $j = 1, 2, \dots, m$, let $d_j = \min\{s(z-1)^j, k\} < n/2$. Let $X_{2i+1} = [0, d_{2i+1} - 1]_n$, $X_{2i} = [n/2, n/2 + d_{2i} - 1]_n$. So $|X_j| = d_j$, and X_{j+1} is an interval of length at most $z|X_j|$. Thus for any vertex x of $K_{p/q}$, $|f(x) \cap X_{j+1}| \leq |X_j \setminus f(x)|$. By Corollary 8, there is a k -fold $(n+s)$ -colouring ϕ of $K_{p/q} \times P_m$ such that $\phi(x, 0) = f(x)$ and $\phi(V(K_{p/q}) \times \{m\})$ uses at most $n+s-k$ colours. By Lemma 1, $\chi_k(\mu_m(K_{p/q})) \leq \chi_k(K_{p/q}) + s = n + s$. ■

The following corollary follows from Corollary 10 and Theorem 11.

Corollary 12 Suppose G is a graph with $\chi_c(G) = \chi_f(G) = p/q$. Let $z = \lfloor p/q \rfloor$. Then for any positive integer s , if $s(z-1)^m \geq k$, then $\chi_k(\mu_m(G)) \leq \chi_k(G) + s$.

4 A technical lemma

In the next section, we shall prove that under certain conditions, we have $\chi_k(\mu_m(G)) = \chi_k(G)$. The proof needs a technical lemma, which we state in this section.

Lemma 13 Let $n = 2s$ be an even integer and $t \geq 1$ be an integer. Suppose for $0 \leq j \leq n-1$ and $0 \leq i \leq t$, $\delta_{i,j}$ are integers satisfying the following equations:

$$\begin{aligned} \delta_{0,0} &= -1 \\ \delta_{0,j} &= 0, \text{ for } j = 1, 2, \dots, n-1 \\ \delta_{i,j} &= \delta_{i-1,[j-1]_n} + \delta_{i-1,j} + \delta_{i-1,[j+1]_n}, \\ &\quad \text{if } i \geq 1 \text{ and } i+j \text{ is even,} \\ \delta_{i,j} &= -\delta_{i-1,j}, \text{ if } i \geq 1 \text{ and } i+j \text{ is odd.} \end{aligned}$$

Then for $i = zs + i'$, where z, i' are integers with $0 \leq i' \leq s-1$, we have

$$\delta_{i,j} = \begin{cases} (-1)^{i+j+1}z, & \text{if } z \text{ is even and } |j|_n > i' \\ & \text{or } z \text{ is odd and } |j|_n < s-i', \\ (-1)^{i+j+1}(z+1), & \text{if } z \text{ is even and } |j|_n \leq i' \\ & \text{or } z \text{ is odd and } |j|_n \geq s-i'. \end{cases}$$

Once the statement of the lemma is understood, the proof of the lemma is just a straightforward but tedious verification (by induction on i). So instead of presenting a proof of the lemma, we give an example of the numbers $\delta_{i,j}$ for $s = 3, t = 9$. The following is the matrix $[\delta_{i,j}]_{t \times n}$ (note that our indices starts with 0. So the first entry is $\delta_{0,0}$):

$$\begin{pmatrix} \mathbf{-1} & 0 & 0 & 0 & 0 & 0 \\ 1 & \mathbf{-1} & 0 & 0 & 0 & \mathbf{-1} \\ -1 & 1 & \mathbf{-1} & 0 & -1 & 1 \\ 1 & -1 & 1 & \mathbf{-2} & 1 & -1 \\ -1 & 1 & -2 & 2 & -2 & 1 \\ 1 & -2 & 2 & -2 & 2 & -2 \\ \mathbf{-3} & 2 & -2 & 2 & -2 & 2 \\ 3 & -3 & 2 & -2 & 2 & -3 \\ -3 & 3 & -3 & 2 & -3 & 3 \\ 3 & -3 & 3 & \mathbf{-4} & 3 & -3 \end{pmatrix}$$

Note that the absolute values of the numbers $\delta_{i,j}$ becomes large and large when i increases. The boldfaced numbers are the “first” places where the numbers $\delta_{i,j}$ have absolute values 1, 2, 3, 4, respectively.

Corollary 14 For any i, j , $\delta_{i,j} + \delta_{i,j+1} \leq 0$.

Corollary 14 follows from the formula for $\delta_{i,j}$ in Lemma 13. It can also be proved easily by induction on i , by using the definition.

5 Graphs with $\chi_k(\mu_m(G)) = \chi_k(G)$

The upper bound $\chi_k(\mu_m(G)) \leq \chi_k(G) + 1$ for sufficiently large m is tight in general. As observed in the introduction, if $\chi_k(G) = k\chi_f(G)$, then for any integer m , $\chi_k(\mu_m(G)) \geq k\chi_f(\mu_m(G)) > k\chi_f(G) = \chi_k(G)$, and hence $\chi_k(\mu_m(G)) \geq \chi_k(G) + 1$. It is natural to ask if $\chi_k(G) > k\chi_f(G)$, then is it true that for sufficiently large m , $\chi_k(\mu_m(G)) = \chi_k(G)$? The answer is no in general. However, for some graphs G , we do have a positive answer. In this section, we present a sufficient condition under which a graph G has $\chi_k(\mu_m(G)) = \chi_k(G)$ for sufficiently large m .

Theorem 15 Suppose $n/k > p/q$, $n = 2s$ is even. Let $t^* = nq - kp \geq 1$, and let $s^* = \lceil q/t^* \rceil$. If $m' \geq s^*s + k$, then

$$\chi_k(\mu_{m'}(K_{p/q})) \leq n.$$

Proof. Let $m = s^*s$ and let $G = K_{p/q} \times P_{m+k}$. In the following, we construct an extendable k -fold n -colouring of G . For convenience, we shall allow a vertex receives more than k colours. Of course, colour sets assigned to adjacent vertices still need to be disjoint.

For $i = 0, 1, \dots, m$, let $V_i = V(K_{p/q}) \times \{i\}$. We colour the sets $V_0, V_1, \dots, V_m$ recursively. First we colour V_0 . Let

$$\begin{aligned} I_0^0 &= [0, q - t^* - 1]_p \\ I_j^0 &= [jq - t^*, (j+1)q - t^* - 1]_p, \\ &\quad \text{for } j = 1, 2, \dots, n-1. \end{aligned}$$

Then we assign colour j to vertex $(x, 0) \in V_0$ if and only if $x \in I_j^0$. We shall regard the vertices in V_0 as a copy of $S(p)$, which is a circle of length p . For each colour j , $I_j^0 \times \{0\}$ is the set of vertices which received colour j , and is called the colour class of colour j . Since each colour class is an interval of $S(p)$ of length at most q , each colour class is an independent set of G . The intervals $I_0^0, I_1^0, \dots, I_{n-1}^0$ are consecutive intervals, with $\sum_{j=0}^{n-1} \ell(I_j^0) = kp$. This means that the union of these intervals winds around the circle $S(p)$ exactly k times. Hence each vertex $(x, 0) \in V_0$ lies in exactly k colour classes.

Therefore we have indeed defined a k -fold colouring of $G[V_0]$.

For $i = 0, 1, \dots, m$, for $j = 0, 1, \dots, n-1$, let $\delta_{i,j}$ be the integers as defined in Lemma 13. Then the intervals I_j^0 has length $q + \delta_{0,j}t^*$, for $j = 0, 1, \dots, n-1$.

Suppose we have defined a k -fold colouring ϕ of $G[V_0 \cup V_1 \cup \dots \cup V_{i-1}]$ and $i < m$. For $0 \leq i' < i$, let

$$I_j^{i'} = \{x \in \{0, 1, \dots, p-1\} : \phi(x, i') = j\}.$$

Suppose further that $0 \leq i \leq m$ and for each $0 \leq j \leq n-1$, the set $I_j^{i'}$ is an interval of $S(p)$ of length $\max\{0, q + \delta_{i,j}t^*\}$. We extend ϕ to the set V_i by defining intervals I_j^i for $j = 0, 1, \dots, n-1$:

If $I_j^{i-1} = \emptyset$, then let I_j^i be the $S(p)$. If $I_j^{i-1} = S(p)$ or $q + \delta_{i,j} \leq 0$, then let $I_j^i = \emptyset$.

Assume $I_j^{i-1} = [a, b]_p$ and $q + \delta_{i,j} > 0$. Then let

$$I_j^i = \begin{cases} [a - \delta_{i-1,j-1}t^*, b + \delta_{i-1,j+1}t^*]_p, & \text{if } i+j \text{ is even,} \\ [a + \delta_{i-1,j}t^*, b - \delta_{i-1,j}t^*]_p, & \text{if } i+j \text{ is odd.} \end{cases}$$

Here the calculation in the second indices are modulo n .

For those non-empty intervals $I_j^i \neq S(p)$, we have

$$\ell(I_j^i) = \begin{cases} \ell(I_j^{i-1}) + \delta_{i-1,j-1}t^* + \delta_{i-1,j+1}t^*, & \text{if } i+j \text{ is even,} \\ \ell(I_j^{i-1}) - 2\delta_{i-1,j}t^*, & \text{if } i+j \text{ is odd.} \end{cases}$$

Therefore

$$\ell(I_j^i) = q + \delta_{i,j}t^*.$$

Now we consider which intervals I_j^i are empty. By the definitions of $t^*, m, q + \delta_{i,j}t^* \leq 0$ if and only if $\delta_{i,j} \leq -s^*$. By referring to Lemma 13, we conclude that

- If $i < m$, then $q + \delta_{i,j}t^* > 0$.
- If s^* is even, then $q + \delta_{m,0}t^* \leq 0$, and $q + \delta_{m,j}t^* > 0$ for $j \neq 0$.
- If s^* is odd then $q + \delta_{m,s^*}t^* \leq 0$, and $q + \delta_{m,j}t^* > 0$ for $j \neq s^*$.

By symmetry, we only consider the case that s^* is even. Again by referring to Lemma 13, it is easy to verify that for $0 \leq i \leq k/2$, we have

- $q + \delta_{m+2i+1,j}t^* \leq 0$ for $j = 2i+1, 2i-1, \dots, 1, n-1, \dots, n-2i-1$.
- $q + \delta_{m+2i,j}t^* \leq 0$ for $j = 2i, 2i-2, \dots, 2, 0, n-2, \dots, n-2i$.

Note that $I_j^i = \emptyset$ means that colour j is not used on vertices in V_i . Thus there are k colours not used on vertices of V_{m+k} . To prove that ϕ is an extendable k -fold n -colouring of G , it suffices to prove the following claim:

Claim 1 *The mapping ϕ defined as above is indeed a k -fold n -colouring of $G[V_0 \cup V_1 \cup \dots \cup V_i]$.*

By our assumption, the restriction of ϕ to $V_0 \cup V_1 \cup \dots \cup V_{i-1}$ is a k -fold colouring of $G[V_0 \cup V_1 \cup \dots \cup V_{i-1}]$. It is straightforward to verify, by using the definition, that if I_j^i and I_{j+1}^i are both non-empty and not equal to $S(p)$, then they are consecutive intervals. If $I_j^i = \emptyset$, then I_{j-1}^i and I_{j+1}^i are either consecutive or at least one of them is equal to $S(p)$. Thus the union $\cup_{j=0}^{n-1} I_j^i$ winds around $S(p)$ k times, with some overlaps at places where $I_j^i = S(p)$. Thus each vertex of V_i is assigned at least k colours. It remains to show that for any $x \in I_j^{i-1}$ and $y \in I_j^i$, xy is not an edge of $K_{p/q}$, i.e., $|x - y|_p < q$.

First we consider the case that $i+j$ is even. Assume $I_j^{i-1} = [a, b]_p$. Then $I_j^i = [a - \delta_{i-1,j-1}t^*, b + \delta_{i-1,j+1}t^*]_p$. For any $x \in I_j^{i-1}$ and $y \in I_j^i$, we have

$$|y - x|_p \leq \max\{ |b + \delta_{i-1,j+1}t^* - a|_p, |b - a + \delta_{i-1,j-1}t^*|_p \}.$$

As $[a, b]_p$ has length $q + \delta_{i-1,j}t^*$, we have

$$|b - a|_p = \delta_{i-1,j}t^* - 1.$$

Thus

$$|y - x|_p \leq \max\{ [q + \delta_{i-1,j}t^* + \delta_{i-1,j+1}t^*]_p - 1, [q + \delta_{i-1,j}t^* + \delta_{i-1,j-1}t^*]_p - 1 \}.$$

By Corollary 14, we have $|y - x|_p \leq q - 1$. Similarly $|x - y|_p \leq q - 1$. So $|x - y|_p \leq q - 1$.

The case $i+j$ is odd can be proved similarly (and is easier). We omit the details. Thus ϕ is indeed a k -fold n -colouring of $G[V_0 \cup V_1 \cup \dots \cup V_{i-1}]$. ■

Theorem 16 *If G is a graph with $\chi_k(G) > k\chi_c(G)$ and $\chi_k(G)$ is even, then for sufficiently large m , $\chi_k(\mu_m(G)) = \chi_k(G)$.*

The $k = 1$ case of Theorem 16 is of special interest.

Corollary 17 *If $\chi(G) > \chi_c(G)$ and $\chi(G)$ is even, then for sufficiently large m ,*

$$\chi(\mu_m(G)) = \chi(G).$$

In [12], the notion that “a graph has chromatic number n for a topological reason” is defined. We shall not define this notion, however, we remark that it is shown in [3, 12] that if G has chromatic number n for a “topological” reason, then $\chi(\mu_m(G)) = \chi(G) + 1$ for any m . Thus the following result of [12] is a corollary of Corollary 17:

Corollary 18 *If G has chromatic number n for a topological reason and n is even, then $\chi(G) = \chi_c(G)$.*

It is proved in [6] that for any positive even number n and any nonnegative integer m , $\chi_c(\mu_m(K_n)) = n + \frac{1}{t}$, where $t = \lfloor 2m/n \rfloor + 1$. Let $G = \mu_m(K_n)$, where $2m \geq n$. Then $\chi(G) > \chi_c(G)$. It is known that G has chromatic number $n + 1$ for a topological reason [12]. Therefore for any integer m' , $\chi(\mu_{m'}(G)) = \chi(G) + 1$. This implies that the condition in Theorem 16 that $\chi_k(G)$ be even is necessary (at least when $k = 1$). However, in some cases, even if $\chi_k(G)$ is odd, the conclusion of Theorem 16 is still true.

Theorem 19 *Assume G is a graph with $\chi_k(G) = n$. Suppose there are positive integers k_1, k_2, n_1, n_2 such that $k = k_1 + k_2$, $n = n_1 + n_2$, n_1 is even, $n_1/k_1 > \chi_c(G)$ and $n_2/k_2 \geq \chi_c(G)$. Then for sufficiently large m' we have $\chi_k(\mu_{m'}(G)) = n$.*

Proof. Assume $\chi_c(G) = p/q$. Then $G \rightarrow K_{p/q}$, and hence $\chi_k(G) \leq \chi_k(K_{p/q})$. Thus it suffices to prove that $\chi_k(\mu_{m'}(K_{p/q})) \leq n$. Assume $n_1 = 2s$, $t^* = qn_1 - k_1p$, $s^* = \lceil q/t^* \rceil$, and $m = s^*s$. Using the proof of Theorem 15, one can prove that there is a k_1 -fold n_1 -colouring ϕ of $K_{p/q} \times P_{m+n_1/2}^*$, using colour set C , such that the set $V_{m+n_1/2}$ uses only half of the colours. Let D be the set of colours not used by vertices in $V_{m+n_1/2}$. As $n_1/2 > k$, we can choose a proper subset B of D of k -colours, and assign it to all the vertices of $V_{m+n_1/2+1}$. Let $c \in C \setminus B$ be a colour not used by any vertices in $V_{m+n_1/2} \cup V_{m+n_1/2+1}$.

Let ψ be a k_2 -fold n_2 -colouring of $K_{p/q}$, using colour set C' . By Theorem 3, ψ can be extended to an extendable k_2 -fold $(n_2 + 1)$ -colouring of $K_{p/q} \times P_{2k_2}$, using colour set $C' \cup \{c\}$. Now we define an extendable k -fold n -colouring f of $K_{p/q} \times P_{m+2k}^*$ as follows: For $x \in V(K_{p/q})$ for $i \leq m$, let $f(x, i) = \phi(x, i) \cup \psi(x, 0)$. For $1 \leq i \leq 2k$, let $f(x, m+i) = \phi(x, m+i) \cup \psi(x, i)$. It is easy to verify that f is an extendable k -fold n -colouring of $K_{p/q} \times P_{m+2k}^*$. ■

As an example of graph that Theorem 15 does not apply but Theorem 19 applies, we consider the

graph $G = K_{13/4}$. We have $\chi_5(G) = 17$. As $\frac{17}{5} = \frac{4+13}{1+4}$, by Theorem 19, we conclude that sufficiently large m , $\chi_5(\mu_m(G)) = 17$. This conclusion cannot be derived from Theorem 15.

Corollary 20 *Suppose $\chi_c(G) = p/q$ and p is odd. Then for any $k \geq q$, if $\chi_k(G) > k\chi_c(G)$, then for sufficiently large m , $\chi_k(\mu_m(G)) = \chi_k(G)$.*

Proof. Assume $\chi_k(G) = n$. If n is even, then the conclusion follows from Theorem 16. If n is odd, then let $n_1 = n - p, n_2 = p, k_1 = k - q$ and $k_2 = q$, the conclusion follows from Theorem 19. ■

References

- [1] J. A. Bondy and P. Hell, *A note on the star chromatic number*, J. Graph Theory **14** (1990), 479-482.
- [2] C.D. Godsil, *Problems in algebraic combinatorics*, Electron. J. Combin. **2**(1995).
- [3] A. Gyárfás, T. Jensen, M. Stiebitz, *On graphs with strongly independent colour-classes*, Journal of Graph Theory, **46** (2004) 1-14.
- [4] G. Hahn and C. Tardif, *Graph homomorphisms: structure and symmetry*, Graph symmetry (Montreal, PQ, 1996), NATO Adv. Sci. Inst. Ser. C Math. Phys. Sci., (497(1997), 107-166.
- [5] P. Hell and J. Nešetřil, *Graph homomorphisms*, Oxford University Press, 2004.
- [6] P.C.B. Lam, W. Lin, G. Gu and Z. Song, *Circular chromatic number and a generalization of construction of Mycielski*, J. Combin. Theor. Ser. B. **89** (2003) 195-205.
- [7] D.D.-F. Liu, *Circular chromatic number for iterated Mycielski graphs*, Discrete Math. **285** (2004) 335-340.
- [8] Lovász, L., *Kneser's conjecture, chromatic number, and homotopy*, J. Comb. Theory (Ser. A) **25** (1978) 319-324.
- [9] M. Larsen, J. Propp and D. H. Ullman, *The fractional chromatic number of Mycielski graphs*, J Graph Theory **19** (1995), 411-416.
- [10] J. Matoušek and G. Ziegler, *Topological lower bounds for the chromatic number: A hierarchy*, Jahresbericht der Deutschen Mathematiker-Vereinigung, **106**(2004), 71-90.

- [11] E. R. Scheinerman, D. H. Ullman, *Fractional Graph Theory*. Wiley-Interscience Series in Discrete Mathematics and Optimization (John Wiley & Son, New York, 1997).
- [12] Gábor Simonyi and Gábor Tardos. Local chromatic number, Ky Fan's theorem, and circular colourings. *Preprint, arXiv:math.CO/0407075*, 2004.
- [13] S. Stahl, n -fold colourings and associated graphs, J. Combin. Theory Ser. B 20 (1976) 185-203.
- [14] C. Tardif, *Fractional chromatic numbers of cones over graphs*, J. Graph Theory 38 (2001) 87-94.
- [15] C. Tardif and X. Zhu, *On Hedetniemi's conjecture and the colour template scheme*, Discrete Math. 253 (2002), no. 1-3, 77-85
- [16] A. Vince, *Star chromatic number*, J. Graph Theory 12 (1988), 551-559.
- [17] X. Zhu, *Circular chromatic number: a survey*, Discrete Math., 229(2001) 371-410.

Proof of Lemma 13

Proof. The proof of the lemma is a straightforward verification by induction on i . For $i = 0$, this follows from the definition. Assume the conclusion for $i - 1$. Assume $i = zs + i'$.

Case 1. z is even and $1 \leq i' \leq s - 1$.

Assume $i + j$ is odd. Since z is even, $i' + j$ is odd, and hence $[j]_n \neq i'$. For $|j|_n > i'$, we have $\delta_{i,j} = -\delta_{i-1,j} = z$. For $|j|_n < i'$, we have $\delta_{i,j} = -\delta_{i-1,j} = z + 1$.

Assume $i + j$ is even. Then $i' + j$ is even, and hence $[j]_n \neq i' - 1$. For $|j|_n > i'$, we have $|j|_n, |j - 1|_n, |j + 1|_n > i' - 1$. Hence

$$\delta_{i,j} = \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} = -z + z - z = -z.$$

For $|j|_n \leq i' - 2$, we have $|j - 1|_n, |j|_n, |j + 1|_n \leq i' - 1$ and hence

$$\begin{aligned} \delta_{i,j} &= \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} \\ &= -(z + 1) + (z + 1) - (z + 1) \\ &= -(z + 1). \end{aligned}$$

For $|j|_n = i'$, we have $\{|j - 1|_n, |j + 1|_n\} = \{i' - 1, i' + 1\}$ and hence

$$\begin{aligned} \delta_{i,j} &= \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} = -(z + 1) + z - z \\ &= -(z + 1). \end{aligned}$$

Case 2 z is even and $i' = 0$. In this case, $i - 1 = (z - 1)s + s - 1$.

Assume $i + j$ is odd. Since $i = zs$ is even, we have $[j]_n \neq 0$. So $|j|_n \geq 1 = s - (s - 1)$, implying that $\delta_{i,j} = -\delta_{i-1,j} = z$.

Assume $i + j$ is even. If $|j|_n \leq 2$, then $|j|_n, |j - 1|_n, |j + 1|_n \geq s - (s - 1)$, and hence

$$\delta_{i,j} = \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} = -z + z - z = -z.$$

If $|j|_n = 1$, then $\{|j - 1|_n, |j + 1|_n\} = \{0, 2\}$, and hence

$$\begin{aligned} \delta_{i,j} &= \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} \\ &= -(z - 1) + (z - 1) - z \\ &= -z. \end{aligned}$$

If $|j|_n = 0$, then $|j - 1|_n = |j + 1|_n = 1$, and hence

$$\begin{aligned} \delta_{i,j} &= \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} = -z + (z - 1) - z \\ &= -(z + 1). \end{aligned}$$

Case 3. z is odd and $1 \leq i' \leq s - 1$.

In this case, $i - 1 = zs + (i' - 1)$. Assume $i + j$ is odd. Then $|j|_n \neq s - i'$. If $|j|_n \geq s - i' + 1$,

then $\delta_{i,j} = -\delta_{i-1,j} = z + 1$. If $|j|_n < s - i'$, then $\delta_{i,j} = -\delta_{i-1,j} = z$.

Next we consider the case that $i + j$ is even. If $|j|_n \geq s - i' + 2$, then $|j|_n, |j - 1|_n, |j + 1|_n \geq s - (i' - 1)$. Hence

$$\begin{aligned}\delta_{i,j} &= \delta_{i-1,|j-1|_n} + \delta_{i-1,j} + \delta_{i-1,|j+1|_n} \\ &= (z + 1) - (z + 1) + (z + 1) \\ &= z + 1.\end{aligned}$$

If $|j|_n < s - i'$, then $|j - 1|_n, |j|_n, |j + 1|_n < s - (i' - 1)$ and hence

$$\delta_{i,j} = \delta_{i-1,|j-1|_n} + \delta_{i-1,j} + \delta_{i-1,|j+1|_n} = z - z + z = z.$$

If $|j|_n = s - i'$, then $\{|j - 1|_n, |j + 1|_n\} = \{s - (i' - 1), s - (i' - 1) + 2\}$ and hence

$$\begin{aligned}\delta_{i,j} &= \delta_{i-1,|j-1|_n} + \delta_{i-1,j} + \delta_{i-1,|j+1|_n} = z - z + (z + 1) \\ &= z + 1.\end{aligned}$$

If $|j|_n = s - i' + 1$, then $\{|j - 1|_n, |j + 1|_n\} = \{s - (i' - 1) - 1, s - (i' - 1) + 1\}$. Hence

$$\begin{aligned}\delta_{i,j} &= \delta_{i-1,|j-1|_n} + \delta_{i-1,j} + \delta_{i-1,|j+1|_n} = z - z + (z + 1) \\ &= z + 1.\end{aligned}$$

Case 4. z is odd and $i' = 0$.

In this case, $i - 1 = (z - 1)s + s - 1$. If $i + j$ is odd, then since z is odd, we have $zs + s$ is even, implying that $|j|_n \leq s - 1$. Therefore, $\delta_{i-1,j} = -z$. So $\delta_{i,j} = -\delta_{i-1,j} = z$.

Assume $i + j$ is even. Since $z - 1$ is even, we have $i + s$ is even. If $|j|_n < s$, then $|j|_n \leq s - 2$, and $|j|_n, |j - 1|_n, |j + 1|_n \leq s - 1$, and hence

$$\delta_{i,j} = \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} = -z + z - z = -z.$$

If $|j|_n = s$, then $|j - 1|_n = |j + 1|_n = s - 1$, and hence

$$\begin{aligned}\delta_{i,j} &= \delta_{i-1,j-1} + \delta_{i-1,j} + \delta_{i-1,j+1} = -z + (z - 1) - z \\ &= -(z + 1).\end{aligned}$$

■